

Database Technologies - Self-Learning

1. Explain EXISTS vs IN performance comparison with an example

```
KD4-98876-Anuradha> SELECT *
-> FROM emp
-> WHERE deptno IN (10, 20, 30);
```

empno	ename	job	mgr	hire	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

14 rows in set (0.189 sec)

```
KD4-98876-Anuradha> SELECT *
-> FROM emp e
-> WHERE EXISTS (
->   SELECT 1
->   FROM dept d
->   WHERE d.deptno = e.deptno
-> );
```

empno	ename	job	mgr	hire	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

14 rows in set (0.583 sec)

Understand ON DELETE CASCADE, ON UPDATE CASCADE. Write emp, dept CREATE TABLE statements and apply CASCADE for all foreign keys.

```
CREATE TABLE emp (
  empno INT PRIMARY KEY,
  ename VARCHAR(50),
  job VARCHAR(50),
  sal DECIMAL(10,2),
```

```
deptno INT,  
FOREIGN KEY (deptno)  
REFERENCES dept(deptno)  
ON DELETE CASCADE  
ON UPDATE CASCADE  
);
```

```
INSERT INTO dept VALUES  
(10, 'ACCOUNTING', 'NEW YORK'),  
(20, 'RESEARCH', 'DALLAS'),  
(30, 'SALES', 'CHICAGO'),  
(40, 'OPERATIONS', 'BOSTON');
```

```
INSERT INTO emp VALUES  
(1, 'SMITH', 'CLERK', 800, 20),  
(2, 'ALLEN', 'SALESMAN', 1600, 30),  
(3, 'JONES', 'MANAGER', 2975, 20),  
(4, 'KING', 'PRESIDENT', 5000, 10);
```

```
ON UPDATE CASCADE  
UPDATE dept  
SET deptno = 50  
WHERE deptno = 10;
```

ON DELETE CASCADE

```
DELETE FROM dept  
WHERE deptno = 20;
```

MongoDb Lab Exercises

1. Practice CRUD operations on Mongo.
 1. Create two collections -- emp and dept.
 2. Insert the emp and dept records.
 3. Increase salary of dept 10 emps by 100.
 4. Display emp name, job, deptno, sal in sorted order of deptno, job, sal (desc) for all emps whose sal more than 1000.
 5. Delete the dept with id 40

2. Learn grouping operations using Mongo aggregation pipeline 1. Display job wise total sal, max sal, min sal, and avg sal for all jobs other than PRESIDENT. 2. Display emp name, dept name, income (sal + comm) in sorted order of their income

```
mongosh> use cdacdb
switched to db cdacdb
cdacdb> db.dept.insertMany([
|   { _id: 10, dname: "ACCOUNTING", loc: "NEW YORK" },
|   { _id: 20, dname: "RESEARCH", loc: "DALLAS" },
|   { _id: 30, dname: "SALES", loc: "CHICAGO" },
|   { _id: 40, dname: "OPERATIONS", loc: "BOSTON" }
| ])
{
  acknowledged: true,
  insertedIds: { '0': 10, '1': 20, '2': 30, '3': 40 }
```

```
cdacdb> db.emp.insertMany([
|   { _id: 7369, ename: "SMITH", job: "CLERK", sal: 800, deptno: 20 },
|   { _id: 7499, ename: "ALLEN", job: "SALESMAN", sal: 1600, deptno: 30 },
|   { _id: 7521, ename: "WARD", job: "SALESMAN", sal: 1250, deptno: 30 },
|   { _id: 7566, ename: "JONES", job: "MANAGER", sal: 2975, deptno: 20 },
|   { _id: 7698, ename: "BLAKE", job: "MANAGER", sal: 2850, deptno: 30 },
|   { _id: 7782, ename: "CLARK", job: "MANAGER", sal: 2450, deptno: 10 },
|   { _id: 7839, ename: "KING", job: "PRESIDENT", sal: 5000, deptno: 10 },
|   { _id: 7934, ename: "MILLER", job: "CLERK", sal: 1300, deptno: 10 }
| ])
{
  acknowledged: true,
  insertedIds: { '0': 7369, '1': 7499, '2': 7521, '3': 7566, '4': 7698, '5': 7782, '6': 7839, '7': 7934 }
```

```
cdacdb> db.dept.find()
{ _id: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
{ _id: 20, dname: 'RESEARCH', loc: 'DALLAS' },
{ _id: 30, dname: 'SALES', loc: 'CHICAGO' },
{ _id: 40, dname: 'OPERATIONS', loc: 'BOSTON' }
```

```

cdacdb> db.emp.updateMany(
|   { deptno: 10 },
|   { $inc: { sal: 100 } }
| )
|
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 3,
|   modifiedCount: 3,
|   upsertedCount: 0
| }
cdacdb> db.emp.find({ deptno: 10 })
[
|   { _id: 7782, ename: 'CLARK', job: 'MANAGER', sal: 2550, deptno: 10 },
|   { _id: 7839, ename: 'KING', job: 'PRESIDENT', sal: 5100, deptno: 10 },
|   { _id: 7934, ename: 'MILLER', job: 'CLERK', sal: 1400, deptno: 10 }
| ]

```

```

cdacdb> db.emp.find({ deptno: 10 })
[
|   { _id: 7782, ename: 'CLARK', job: 'MANAGER', sal: 2550, deptno: 10 },
|   { _id: 7839, ename: 'KING', job: 'PRESIDENT', sal: 5100, deptno: 10 },
|   { _id: 7934, ename: 'MILLER', job: 'CLERK', sal: 1400, deptno: 10 }
| ]
cdacdb> db.dept.deleteOne({ _id: 40 })
| { acknowledged: true, deletedCount: 1 }
cdacdb> db.dept.find()
[
|   { _id: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
|   { _id: 20, dname: 'RESEARCH', loc: 'DALLAS' },
|   { _id: 30, dname: 'SALES', loc: 'CHICAGO' }
| ]
cdacdb> db.emp.aggregate([
|   {
|     $match: {
|       job: { $ne: "PRESIDENT" }
|     },
|   },
|   {
|     $group: {
|       _id: "$job",
|       total_sal: { $sum: "$sal" },
|       max_sal: { $max: "$sal" },
|       min_sal: { $min: "$sal" },
|       avg_sal: { $avg: "$sal" }
|     }
|   }
| ])

```

```
_id: 'SALESMAN',  
total_sal: 2850,  
max_sal: 1600,  
min_sal: 1250,  
avg_sal: 1425  
,  
_id: 'CLERK',  
total_sal: 2200,  
max_sal: 1400,  
min_sal: 800,  
avg_sal: 1100  
,  
_id: 'MANAGER',  
total_sal: 8375,  
max_sal: 2975,  
min_sal: 2550,  
avg_sal: 2791.6666666666665
```

```

    db.emp.aggregate([
      {
        $lookup: {
          from: "dept",
          localField: "deptno",
          foreignField: "_id",
          as: "department"
        }
      },
      {
        $unwind: "$department"
      },
      {
        $project: {
          _id: 0,
          ename: 1,
          dname: "$department.dname",
          income: {
            $add: [
              "$sal",
              { $ifNull: ["$comm", 0] }
            ]
          }
        }
      },
      {
        $sort: {
          income: 1
        }
      }
    ])

```

```

[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'WARD', dname: 'SALES', income: 1250 },
  { ename: 'MILLER', dname: 'ACCOUNTING', income: 1400 },
  { ename: 'ALLEN', dname: 'SALES', income: 1600 },
  { ename: 'CLARK', dname: 'ACCOUNTING', income: 2550 },
  { ename: 'BLAKE', dname: 'SALES', income: 2850 },
  { ename: 'JONES', dname: 'RESEARCH', income: 2975 },
  { ename: 'KING', dname: 'ACCOUNTING', income: 5100 }
]

```